Chapter-6

Pattern Recognition in Medical Imaging

2026-01-05

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Medical imaging

- Medical imaging is the technique and process of *imaging the interior of a body* for clinical analysis and medical intervention, as well as visual representation of the function of some organs or tissues (physiology).
- These images are used for a *variety of purposes*, including diagnosing diseases, monitoring treatment progress, and guiding surgical procedures.

X-ray medical imaging

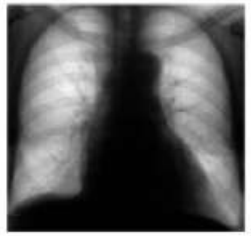
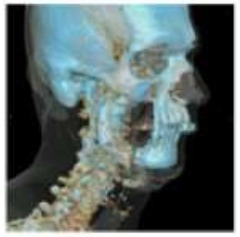
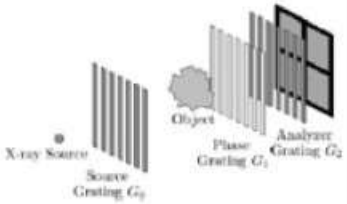
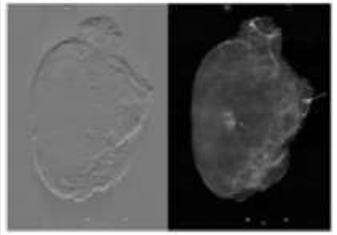
Computed tomography (CT) medical imaging

Ultrasound medical imaging

X-ray transmission imaging

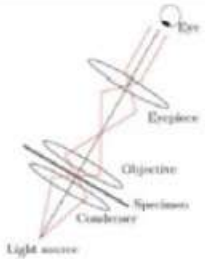
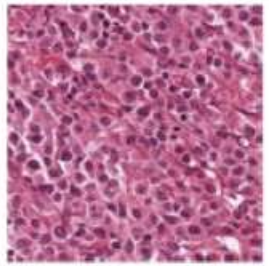
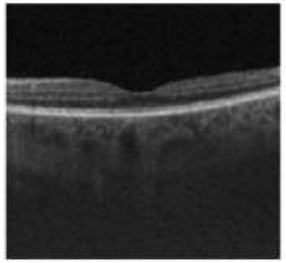
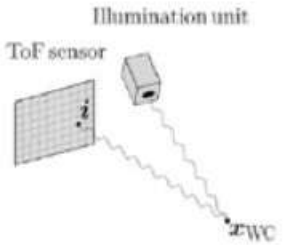
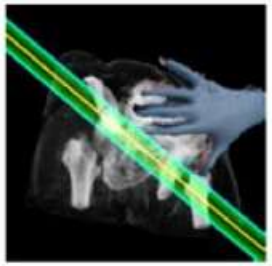
- All of them use high-energy ***X-rays*** that are able to *penetrate soft and hard tissues* in the body.
- Emitted from an X-ray source, the photons pass the body and are collected on the other side by a detector.
- Using the energy loss from source to detector, contrast is generated. The more energy is lost, the higher the contrast.

Data acquisition: The process of capturing the medical image using specialized imaging devices.

Name: Digital radiography **Energy range:** 50–150 kVp **Weight:** 200 kg **Max resolution:** 144 µm (@Detector) **Established in:** 1895		
Name: Mammography **Energy range:** 20–50 kVp **Weight:** 350 kg (stand + generator) **Max resolution:** 85 µm (@Detector) **Established in:** 1927		
Name: C-arm angiography **Energy range:** 70–120 kVp **Weight:** 665 kg (floor stand) **Max resolution:** 150 µm (@Detector) **Established in:** 1954		
Name: Computed tomography **Energy range:** 70–150 kVp **Weight:** 2.5 tons (gantry + couch) **Max resolution:** 450 µm (@Detector) **Established in:** 1971		
Name: X-ray phase contrast and dark-field **Energy range:** 40–100 kVp **Weight:** N.A. – research installation only **Max resolution:** 150 µm (@Detector) **Established in:** 2006		

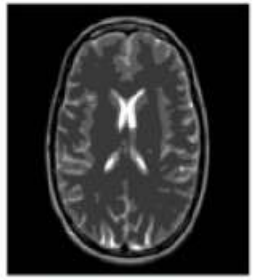
Optical imaging

- All optical modalities operate either within the visible spectrum or at an infrared range that is close to the visible spectrum.
- One of the earliest approaches to explore the inner workings of human beings is endoscopy.

Name: Endoscopy **Physical effect:** Visible light **Weight:** 4.9 kg **Wavelengths:** 400–700 nm **Established in:** 1853		
Name: Microscopy **Physical effect:** Visible light **Weight:** 6 kg **Wavelengths:** 400–700 nm **Established in:** 1609		
Name: Fundus imaging **Physical effect:** Visible light **Weight:** 24.5 kg **Wavelengths:** 400–700 nm **Established in:** 1851		
Name: Optical coherence tomography (OCT) **Physical effect:** Infrared light **Weight:** 65 kg **Wavelengths:** 800–1400 nm **Established in:** 1991		
Name: Depth imaging **Physical effect:** Infrared light **Weight:** 180 g **Wavelengths:** 800–1600 nm **Established in:** 2009		

Large wavelengths and transversal waves

- **Employ long wavelengths and transversal waves.**
- **In MRI**, a strong external magnetic field is used to align dipoles (H_2O used) within the human body.
- **Ultrasound** imaging uses a transceiver to emit and measure pressure waves into bodily tissues.
- Photoacoustic imaging relies on the same transceiver technology as ultrasound imaging for measuring pressure waves.

Name: Magnetic resonance imaging **Physical effect:** Radio waves and magnetic resonance effect **Weight:** 7.3 tons **Wavelengths:** 1–25 m **Established in:** 1974		
Name: Ultrasound **Physical effect:** Sound waves **Weight:** 10 kg **Wavelengths:** 35–400 µm **Established in:** 1942		
Name: Photoacoustic imaging **Physical effect:** Photoacoustic effect **Weight:** 10 kg **Wavelengths:** 300 µm **Established in:** 1993		

Data preprocessing

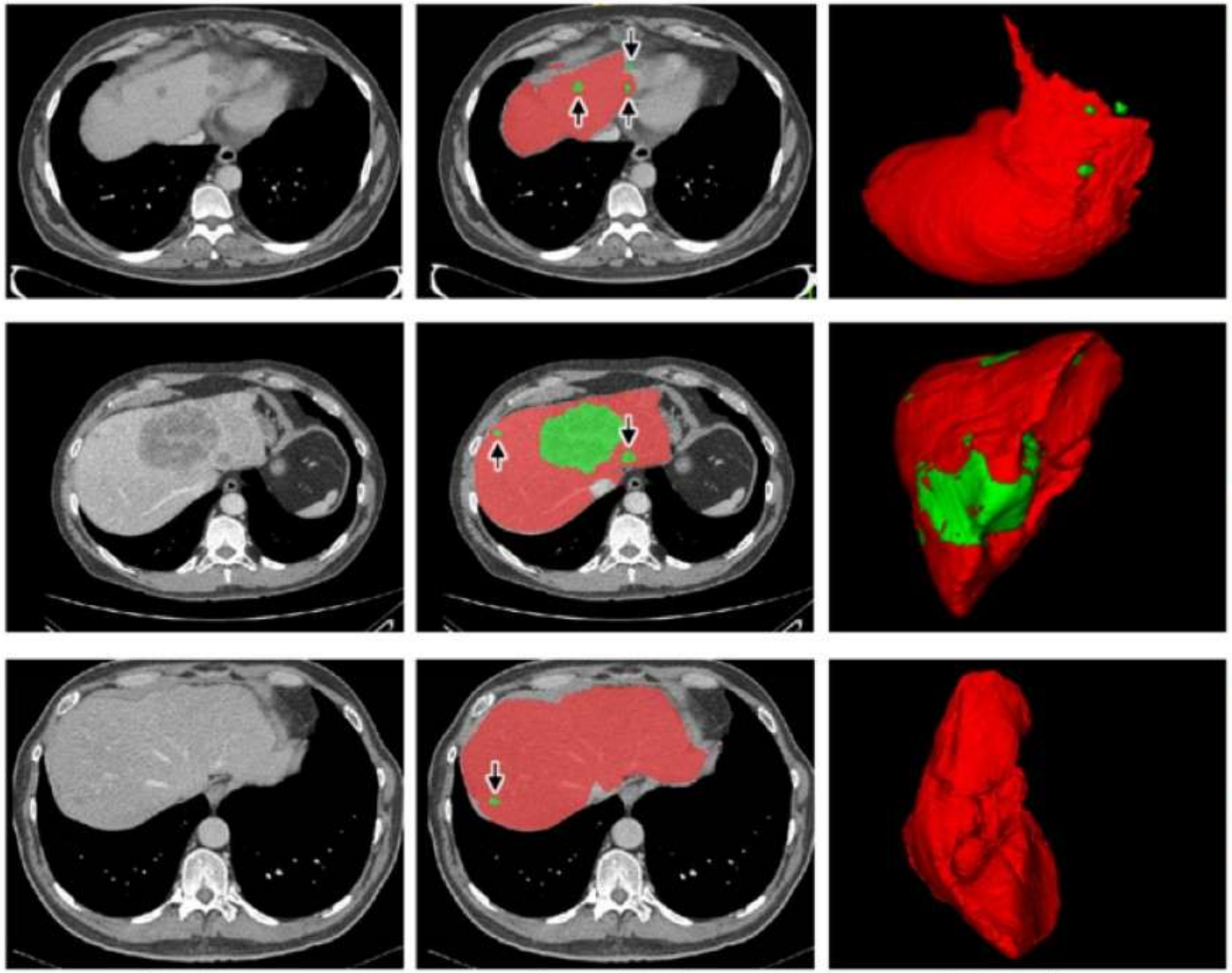
- The raw medical image data often undergoes preprocessing steps to enhance its quality and prepare it for further analysis.
- Preprocessing techniques can address issues like:
  - **Noise Reduction:** Removing or reducing noise from the image through various filtering techniques such as Gaussian filtering or median filtering.
  - **Image adjustment:** Aligning multiple images from different modalities or time points by adjustment in position, orientation, and scale.
  - **Contrast* Enhancement:** Adjusting the contrast of the image like histogram equalization or contrast stretching.
  - **Intensity Normalization:** Ensuring consistent intensity levels across images
  - **Artifact Removal:** Eliminating artifacts caused by factors such as motion, scanner imperfections, or patient-related factors.
  - **Segmentation:** Partitioning the image into meaningful regions or structures of interest.
  - **Feature Extraction:** Quantifying relevant features or characteristics from the image data, such as texture, shape, or intensity patterns.
  - **Spatial Filtering:** Enhancing or suppressing specific spatial frequencies in the image like edge enhancement or smoothing
  - **Image Resampling:** Adjusting the spatial resolution or size of the image using interpolation or decimation of pixel to match the requirements
  - **Quality Control:** Assessing the overall quality and integrity of the pre-processed images using visual inspection, quantitative metrics, or comparison with reference standards.

***Image Contrast** is the degree of variation between the **darkest and brightest regions** of an image i.e. Contrast=$I_{max} - I_{min}$, where I is pixel intensity.

Representation

- Medical image representation involves *converting raw image data into a format* that is suitable for analysis, interpretation, and storage. This representation can involve:
  - **Pixel intensity values:** the numerical representation of the brightness or intensity of each pixel in the image.
  - **Volumetric Representation:** Medical images often consist of a series of 2D slices acquired along different planes (e.g., axial, sagittal, coronal). Volumetric representation involves stacking these 2D slices to create a three-dimensional (3D) volume.
  - **Multi-modal Representation:** Many medical imaging studies involve acquiring images using multiple modalities to capture complementary information about the same anatomical region. Multi-modal representation integrates data from different imaging modalities into a unified framework, enabling comprehensive analysis and interpretation.
  - **Feature-based Representation:** Focuses on extracting and representing specific features or descriptors that characterize relevant anatomical or pathological structures. Features may include texture patterns, shape descriptors, intensity gradients, or local image statistics.
  - **Hierarchical Representation:** Organizes medical image data into multiple levels of abstraction or resolution. For example, multi-resolution analysis by pyramid representation, where coarse representations are used for global context, while finer representations capture detailed information.
  - **Graph-based Representation:** Graph nodes represent image elements, while edges encode spatial or intensity-based relationships between them.
  - **Statistical Representation:** Statistical models such as histograms, probability density functions, and parametric models (e.g., Gaussian mixture models) are used to describe the statistical properties of image data.
  - **Deep Learning Representation:** Deep learning models can capture complex patterns, spatial relationships, and hierarchical features, enabling state-of-the-art performance in tasks

Liver and Tumor Segmentation



Raw image | Ground truth | 3D display

Challenges:

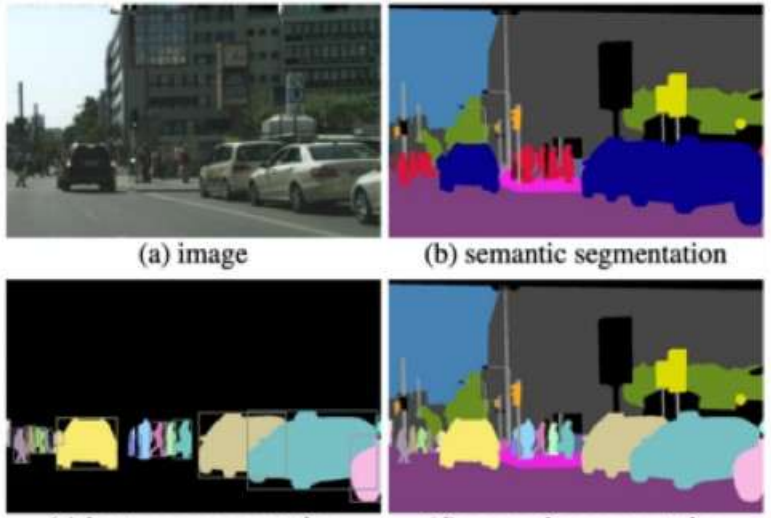
- The *size*, *shape*, and *location* are complex and vary from person to person
- The tumor edge is *blurred* and may be adjacent to other organs and blood vessels
- They are exhibit *grayscale* values *similar* to that of neighboring organs
- There are *pathological changes*
- The current neural network model has *a large number of parameters and low robustness*

Opportunities:

- Deep learning techniques are rapidly developing
- We are in the era of *big data in healthcare*
- Clear *anatomical contextual information* from abdominal CT
- The *powerful computing* capability can be provided

Segmentation (Semantic, Instance and Panoptic)

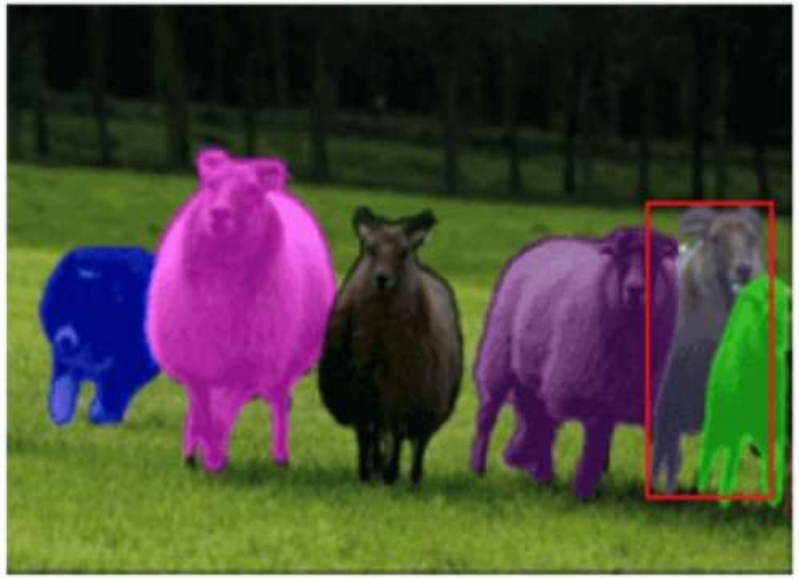
Panoptic segmentation output



(a) image (b) semantic segmentation

(c) instance segmentation (d) panoptic segmentation

Instance segmentation output



Semantic Segmentation

http://www.cs.cornell.edu/courses/cs6670/2021fa/lec08-segmentation.pdf

Semantic Segmentation vs. Instance Segmentation vs. Panoptic Segmentation

(a) Image

(b) Semantic Segmentation

(c) Instance Segmentation

(d) Panoptic Segmentation

https://www.v7labs.com/blog/image-segmentation-guide

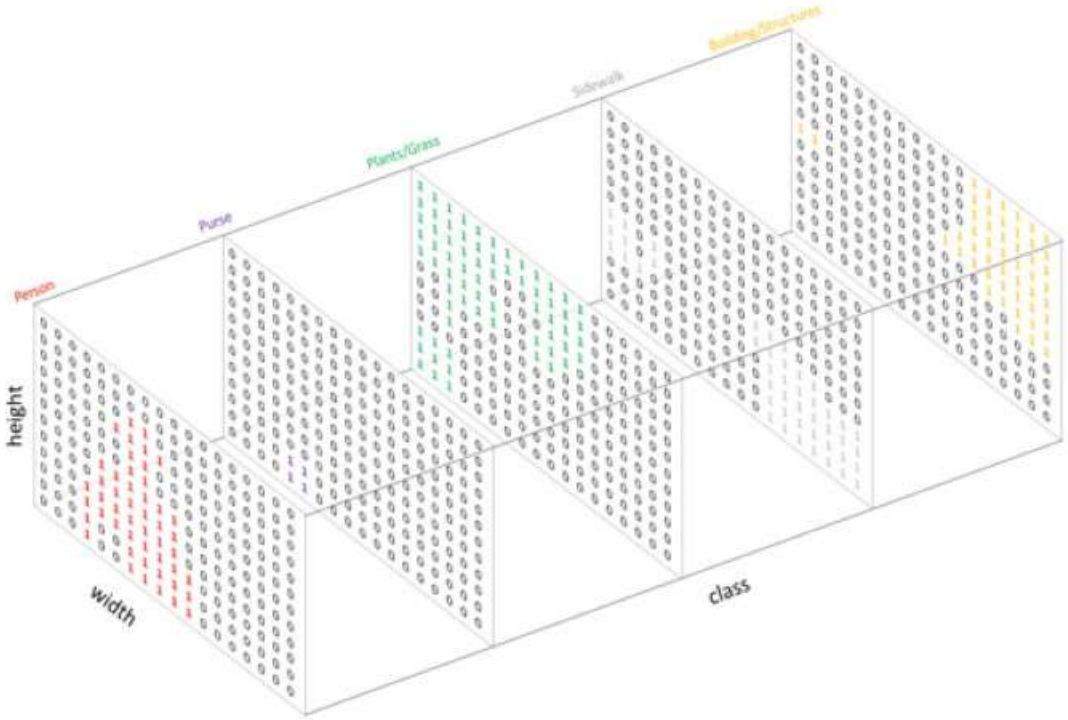
segmented

3	3	3	3	3	3	3	3	3	3	3	3	5	5	5	5	5	5
3	3	3	3	3	3	3	3	3	3	3	3	5	5	5	5	5	5
3	3	3	3	3	3	1	1	3	3	3	3	5	5	5	5	5	5
3	3	3	3	3	1	1	1	1	3	3	3	5	5	5	5	5	5
3	3	3	3	3	3	1	1	3	3	3	5	5	5	5	5	5	5
5	5	3	3	3	3	1	1	3	3	5	5	5	5	5	5	5	5
4	4	3	4	1	1	1	1	1	1	4	4	4	5	5	5	5	5
4	4	3	4	1	1	1	1	1	1	4	4	4	4	4	5	5	5
4	4	4	1	1	1	1	1	1	1	1	4	4	4	4	4	4	4
3	3	3	1	1	1	1	1	1	1	1	4	4	4	4	4	4	4
3	3	3	1	2	2	1	1	1	1	1	4	4	4	4	4	4	4
3	3	3	1	2	2	1	1	1	1	1	4	4	4	4	4	4	4

1: Person
2: Purse
3: Plants/Grass
4: Sidewalk
5: Building/Structures

Fig. Input image and corresponding semantic labels

0: Background/Unknown
1: Person
2: Purse
3: Plants/Grass
4: Sidewalk
5: Building/Structures



Survey Paper: Image Segmentation Using Deep Learning: A Survey

https://arxiv.org/pdf/2001.05566.pdf

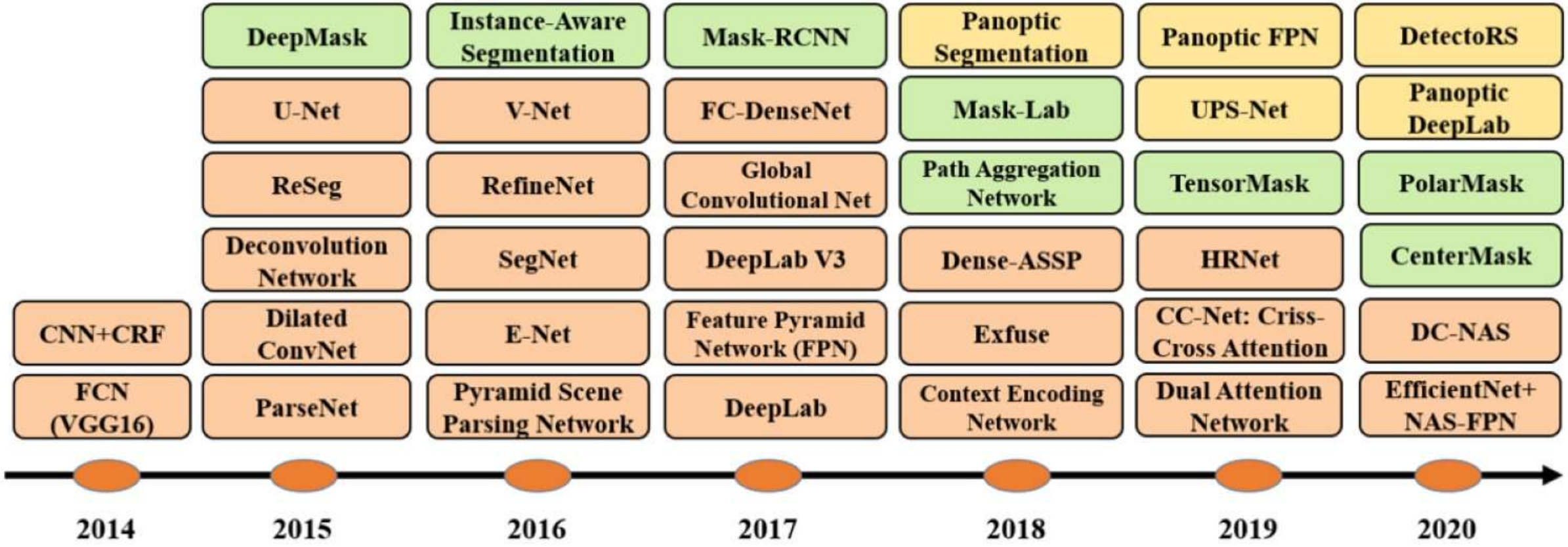
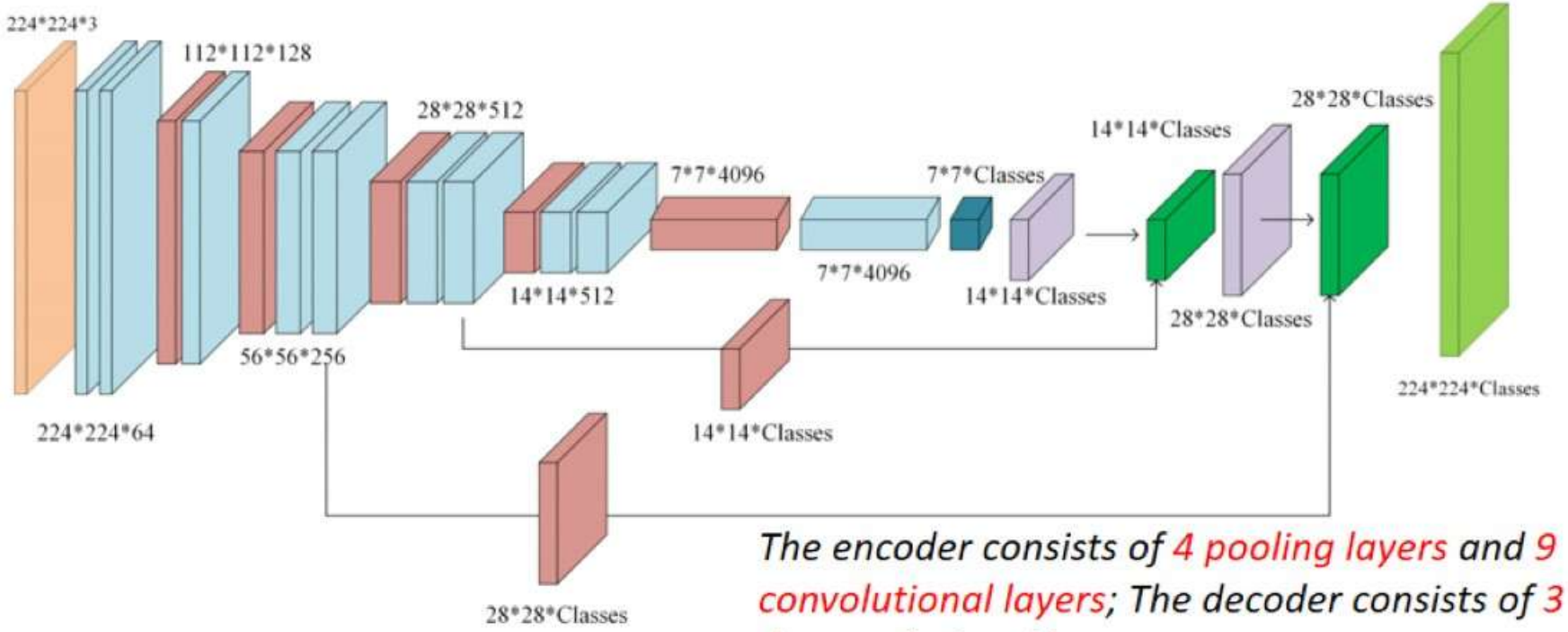


Figure. The timeline of DL-based segmentation algorithms for 2D images, from 2014 to 2020. Orange, green, and yellow blocks refer to semantic, instance, and panoptic segmentation algorithms respectively

FCN (Fully Convolutional Networks)

Disadvantages:

- The *decoder* is too coarse.
- Feature fusion uses the way of *adding*.
- Backbone network is VGG16/19 leading to large memory usage.



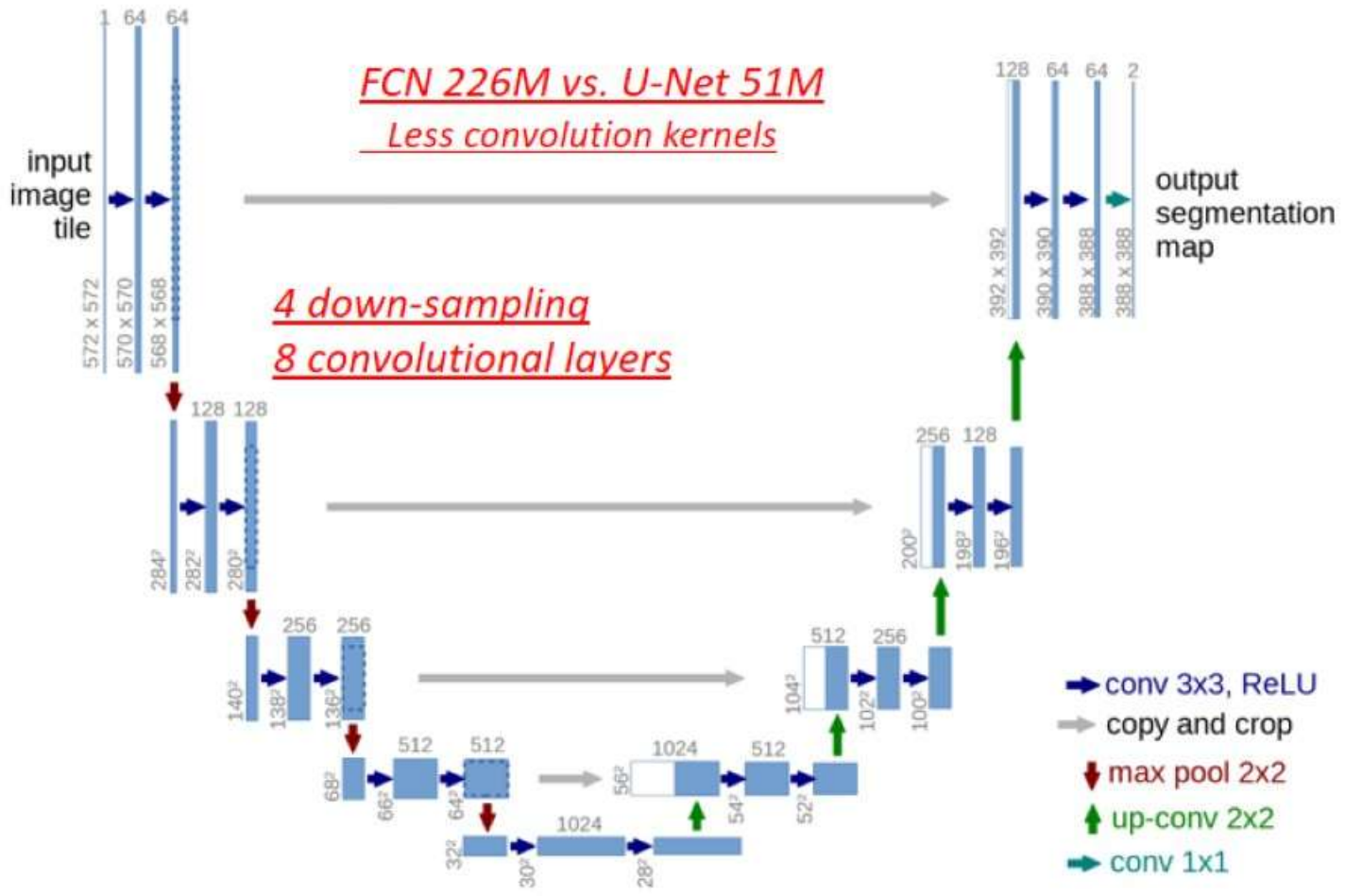
The encoder consists of 4 pooling layers and 9 convolutional layers; The decoder consists of 3 deconvolutional layers.

Advantages:

- FCN firstly presents the idea of *encoder-decoder*, the idea is effective for image segmentation.
- FCN firstly adopts *fully convolution* operation instead of fully connection, which improves the computational efficient.
- FCN firstly adopts the idea of *feature fusion* of low-level and high-level features.

[1] E. Shelhamer, J. Long, T. Darrell, "Fully Convolutional Networks for Semantic Segmentation", IEEE Trans. Pattern Anal. Mach. Intell, vol. 39, no. 4, pp. 640-651, May. 2016.

U-Net



Advantages:

- A *perfectly symmetric* network structure.
- Deconvolution+convolution can improve *feature fusion.*
- Skip connection uses the way of *concatenate*.

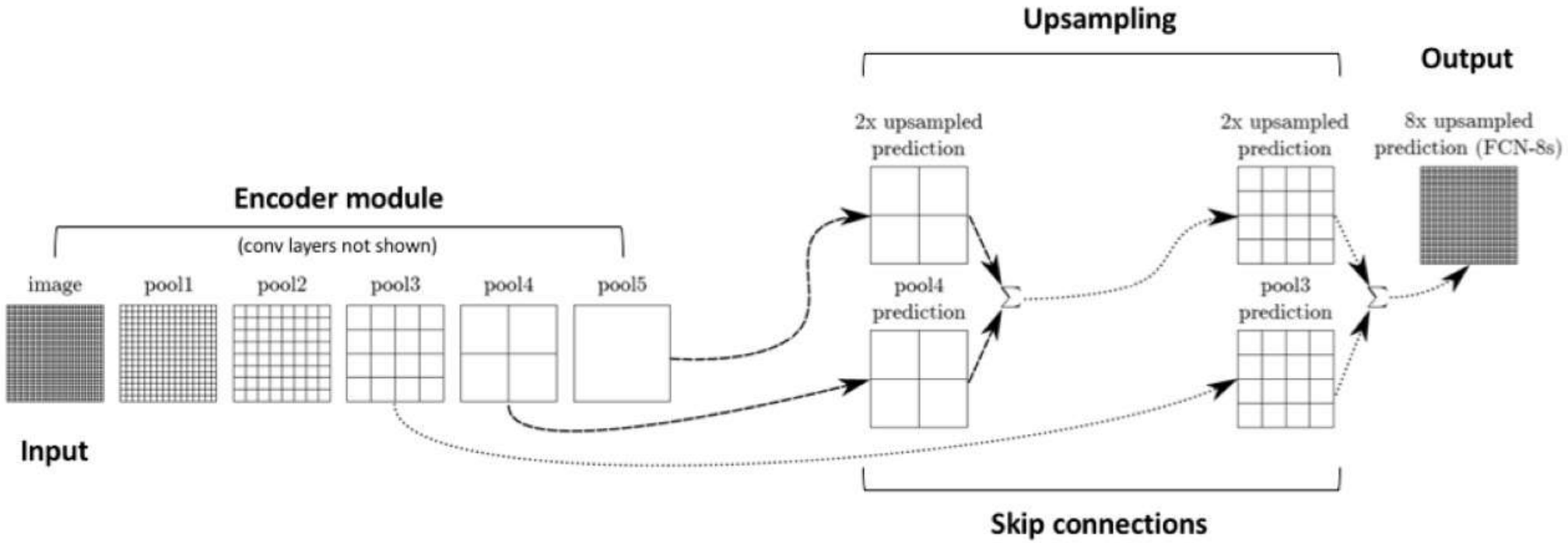
Disadvantages:

- The common convolution adopts convolution kernels with *fixed geometric structures*.
- The *fixed perceptive field* lose the global information
- Skip connection is simple and can be improved.
- The model size is large

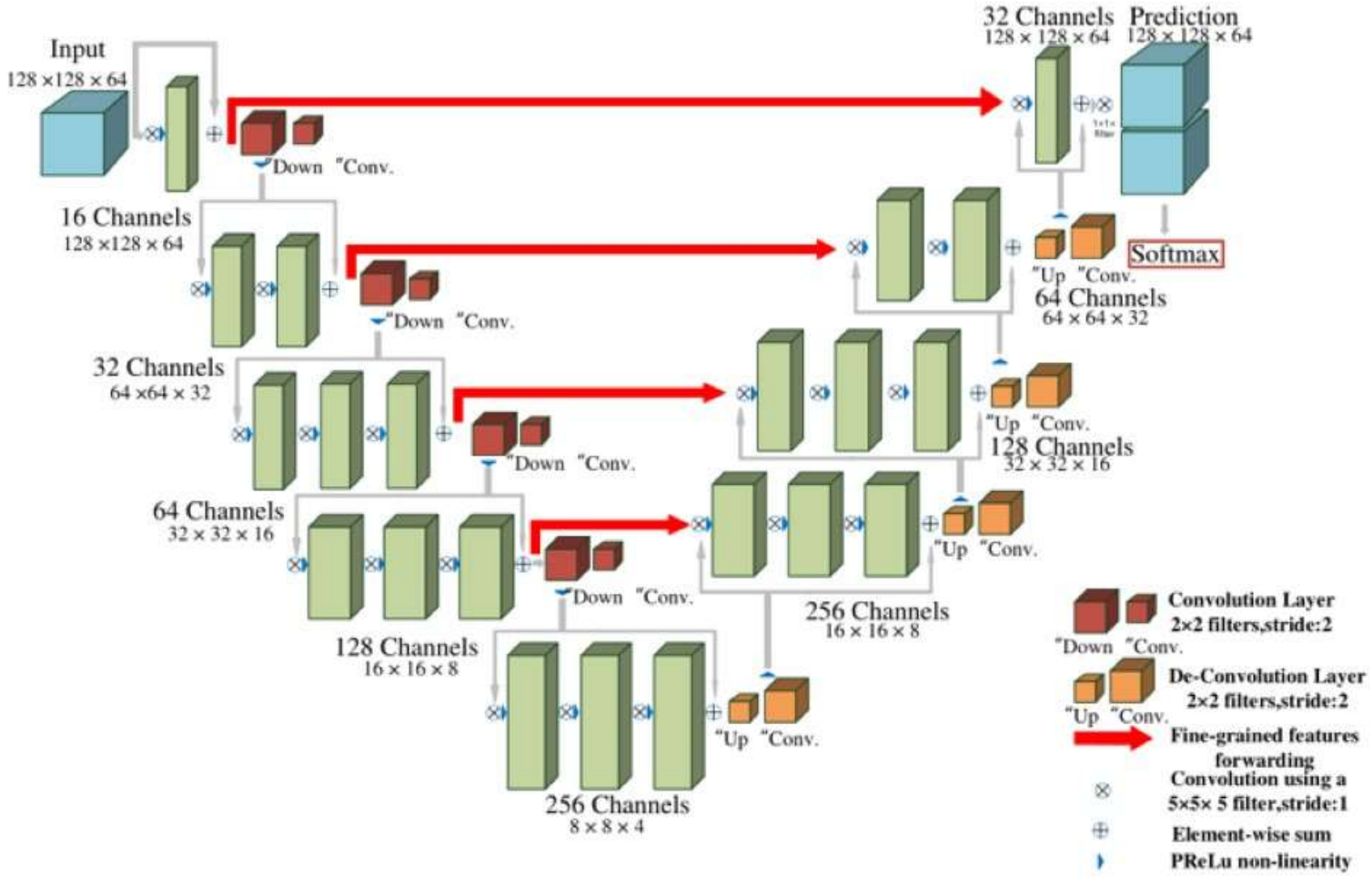
[1] Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," Proc. Int. Conf. Med. Image Comput. Comput. Assist. Intervent. (MICCAI), pp. 234–241, 2015.

Encoder and decoder layer correlation

- Adding "skip connections" from earlier layers, and summing these two feature maps.



V-Net



Advantages:

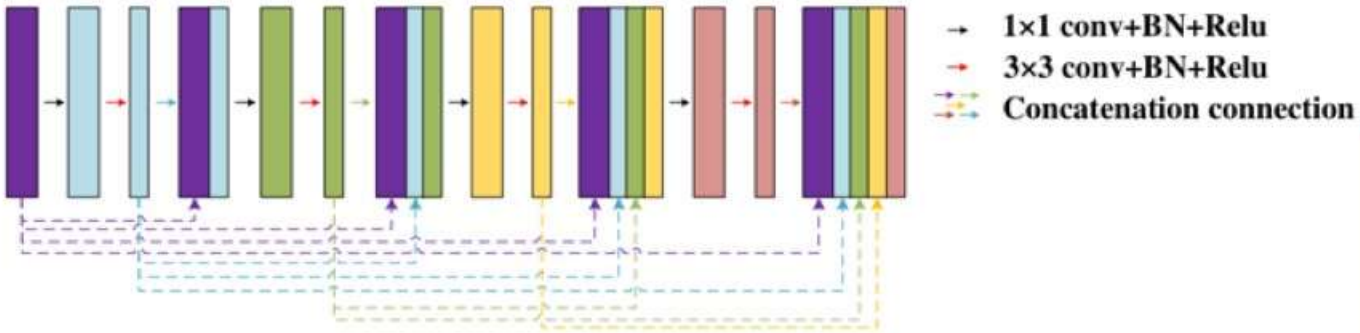
- **3D convolution kernels** are employed to extract context information of *3D space*.
- Residual structure is adopted in the V-Net.

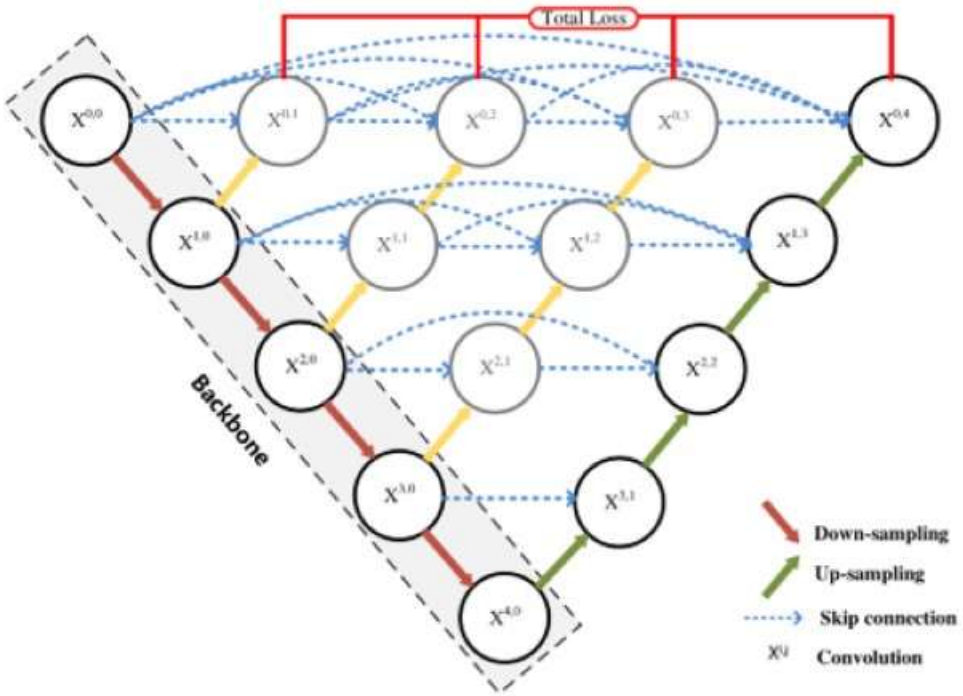
Disadvantages:

- The size of the network is huge due to *a large number of parameters*.
- It requires large number of hardware resources.

[1] F. Milletari, N. Navab, and S. A. Ahmadi, "V-net: Fully convolutional neural networks for volumetric medical image segmentation," Conf. 3D Vis. (3DV) , pp. 565-571, 2016.

Dense U-Net & U-Net++ (Dense Connection Net)





Advantages:

- Dense connection employs *"concat"* that achieves feature reuse.
- Dense connection employing *bottleneck module* to reduce the dimension of feature maps.
- Dense connection has a *low computational complex* since the former is trained easily and achieves convergence fast.
- Dense connection achieves a *hidden depth supervision* due to the error is directly input into each layer

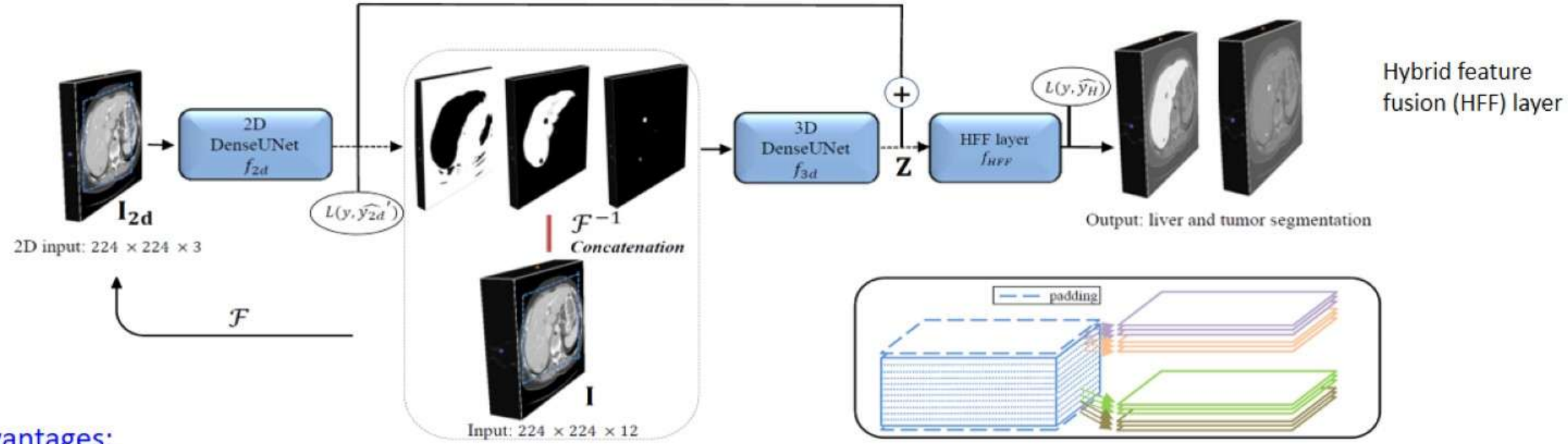
Disadvantages:

- It often *reduces the robustness* of feature representation to a certain extent and *increases the number of parameters.*

[1] S. Guan, A. Khan, S. Sikdar, and P . Chitnis, "Fully dense UNet for 2D sparse photoacoustic tomography artifact removal," IEEE J. Biomed. Health. Inform., 2019.
[2] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: Redesigning skip connections to exploit multiscale features in image segmentation," IEEE Trans. Med. Imag., 2019.

H-DenseUnet

The champion in the challenge of MICCAI2017.



Advantages:

- 2D+3D stronger feature learning capability.
- *Densely connected on U-Net*.
- It converges quickly.

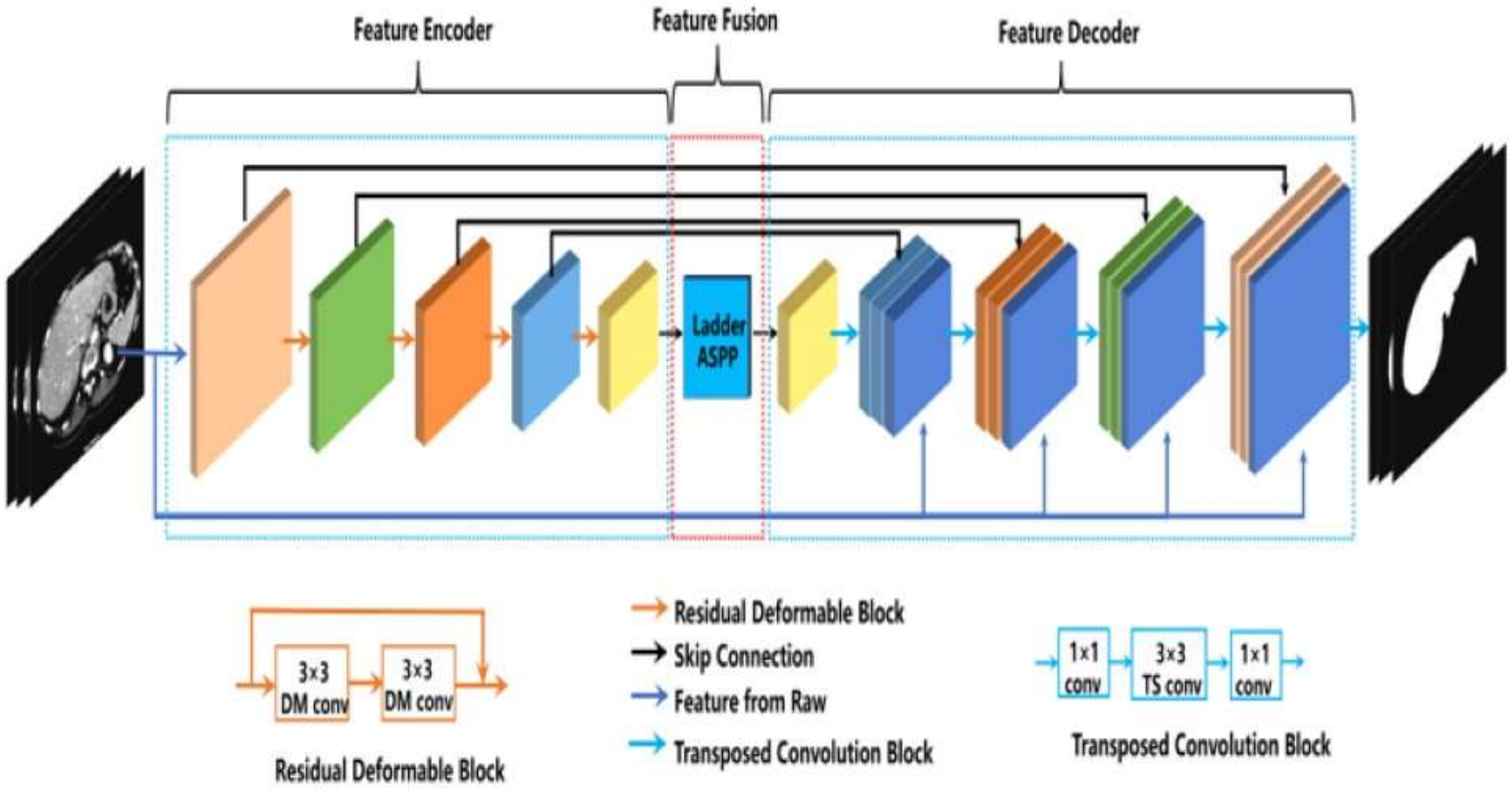
Disadvantages:

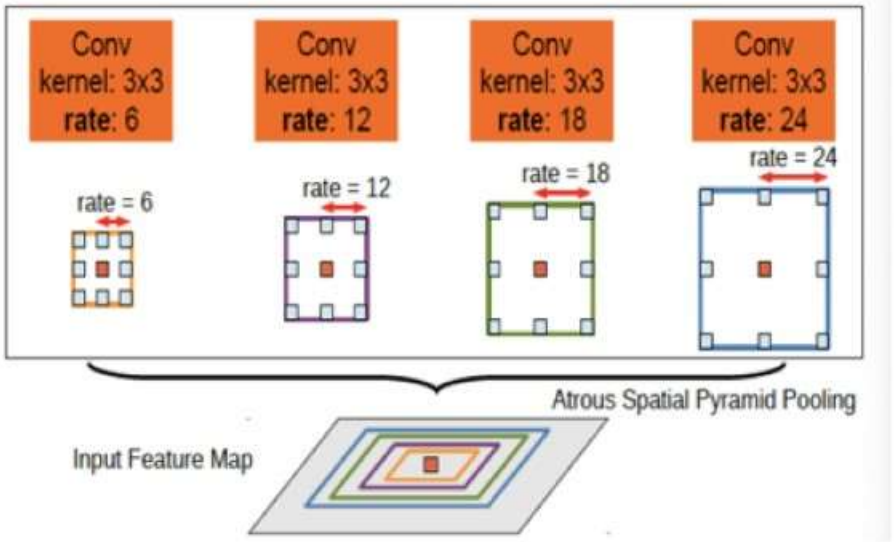
- The network structure is complex, and the training process is very difficult.

Each 3D input volume is sliced into adjacent slices through transformation process F and then fed into 2D DenseUNet; Concatenated with the prediction volumes from 2D network, the 3D input volumes are fed into the 3D network for learning inter-slice features; Then, the HFF layer fused and optimized the intra-slice and inter-slice features for accurate liver and tumor segmentation.

[1] X. Li, H. Chen, X. Qi, Q. Dou, C. Fu, P. Heng, "H-DenseUNet: Hybrid Densely Connected UNet for Liver and Tumor Segmentation From CT Volumes" , IEEE Trans. Med. Imag., vol. 37, no. 12, pp. 2663-2674, Dec. 2018.

DefED-Net



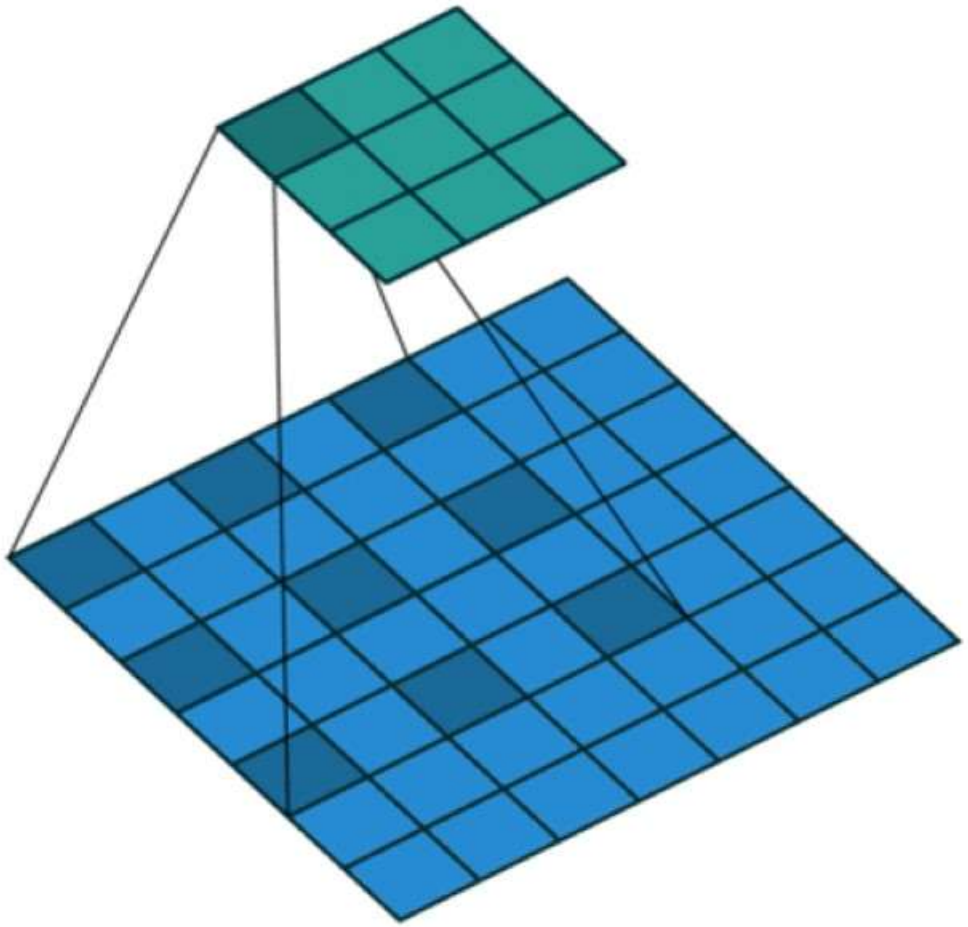


ASPP: Atrous Spatial Pyramid Pooling

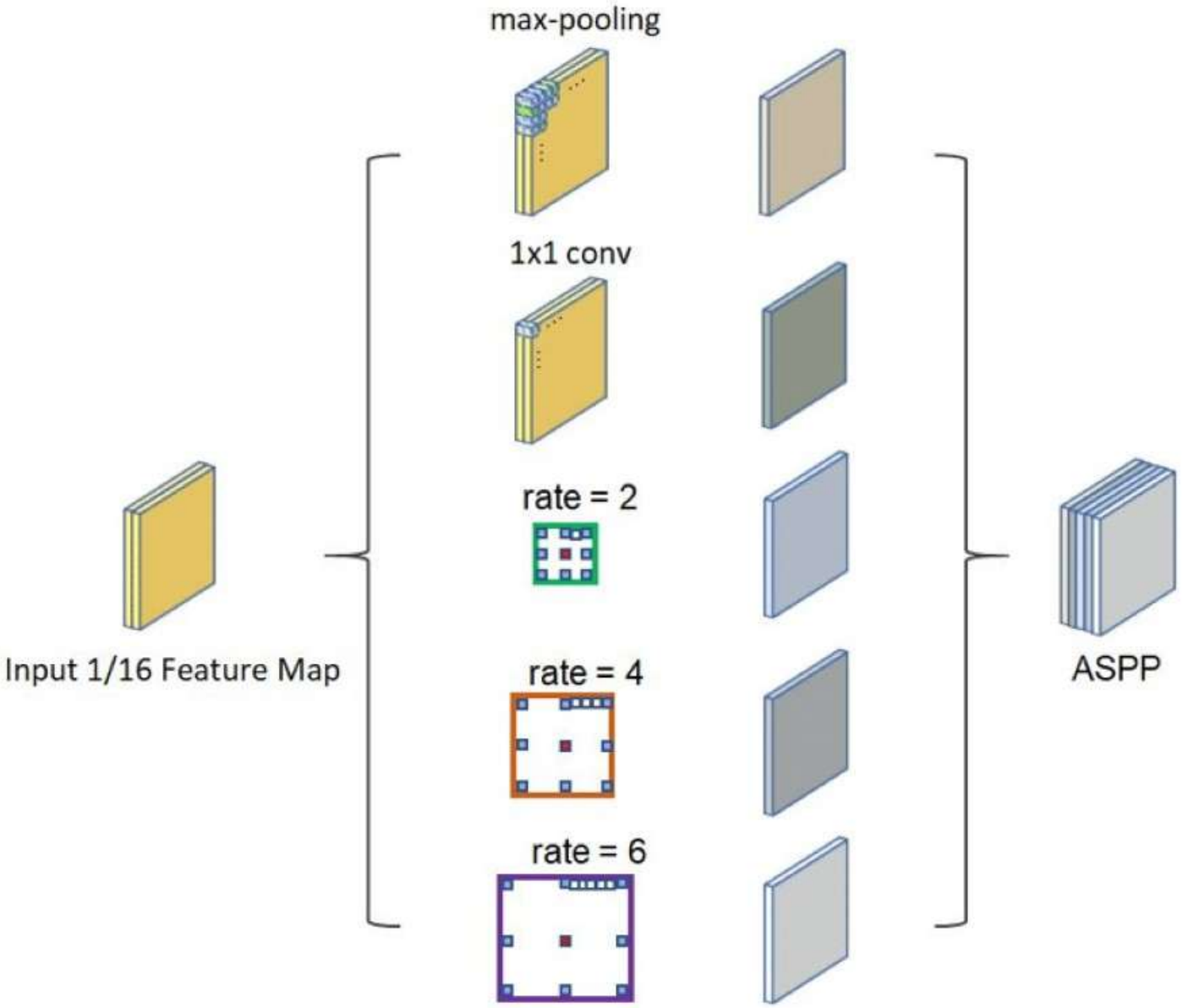
Contributions:

- 1. Feature extraction of CT images using *deformable convolution* with residual connection.
- 2. *Ladder-ASPP* is presented for context information fusion.
- 3. Higher segmentation accuracy for networks with smaller model parameters.

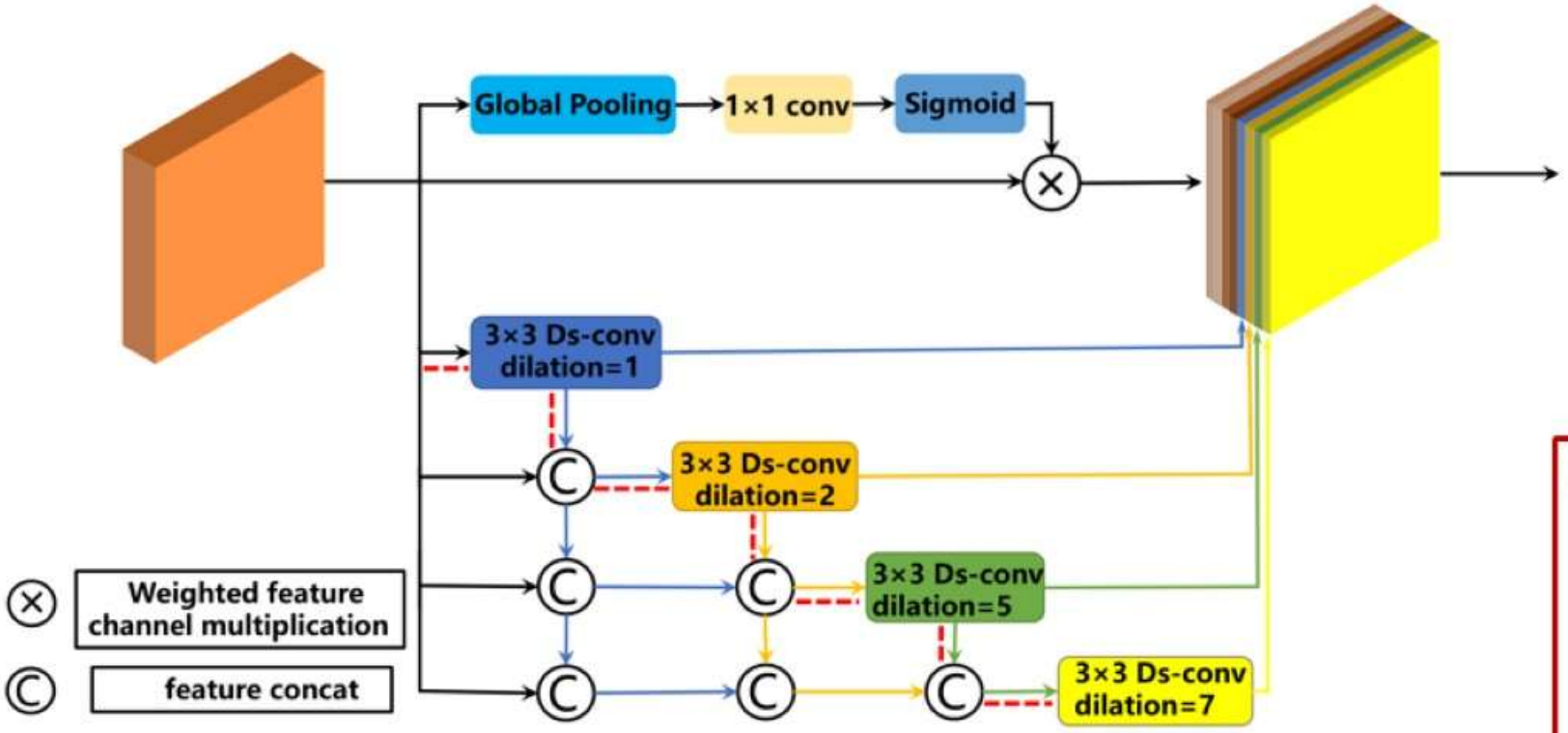
ASPP: Atrous Spatial Pyramid Pooling



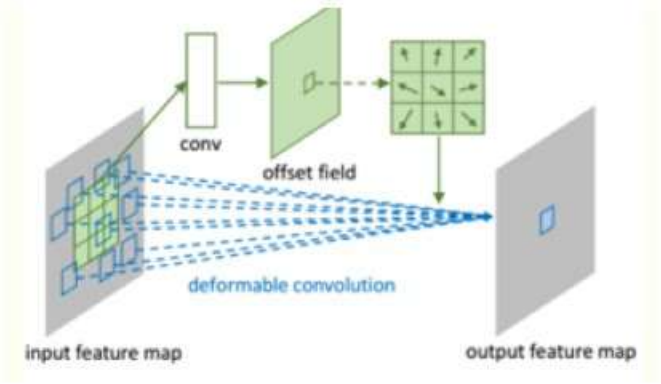
Dilated Convolution (Dilation Rate = 2)



Motivation 2-Ladder ASPP



The block of Ladder-ASPP

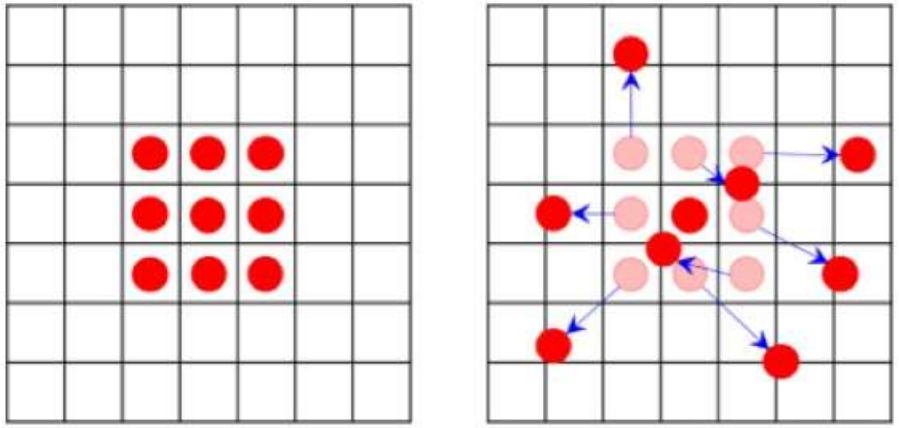


Contributions:

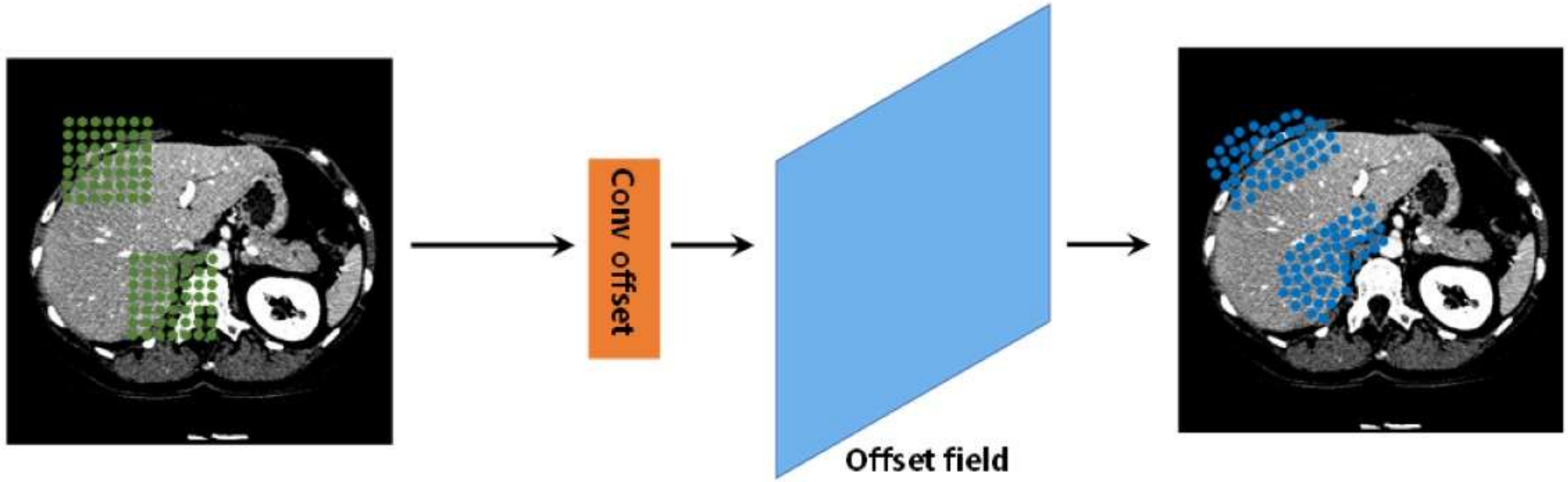
- 1.*Densely connected* spatial pyramid pooling.
- 2. Separate convolution is used for Ladder-ASPP to achieve model compression.
- *Channel attention mechanism is ued for Ladder-ASPP.*

Atrous Spatial Pyramid Pooling (**ASPP**)

Motivation 1-Deformable convolution



- Advantages: *wider receptive field*, *adaptive spatial information.*
- Disadvantages: high computational cost, small increase of the number of parameters.



Deformable convolutions

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- Deformable convolution uses the back-propagation error to automatically adjust the shape of the convolution kernel, thus allowing for adaptive localization and accurate feature extraction for objects of different shapes and sizes, with the disadvantage that a small part of the features will be lost.

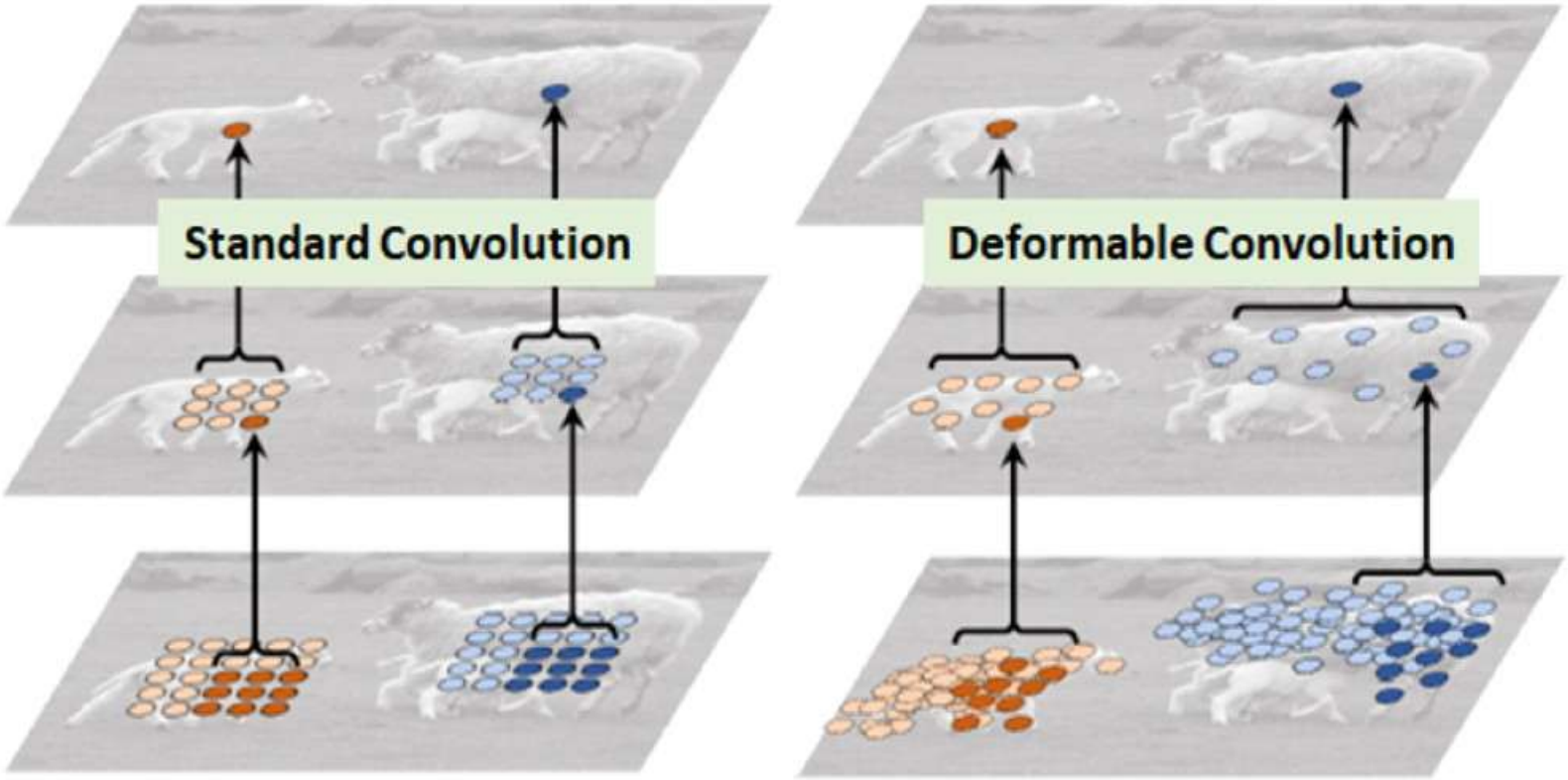
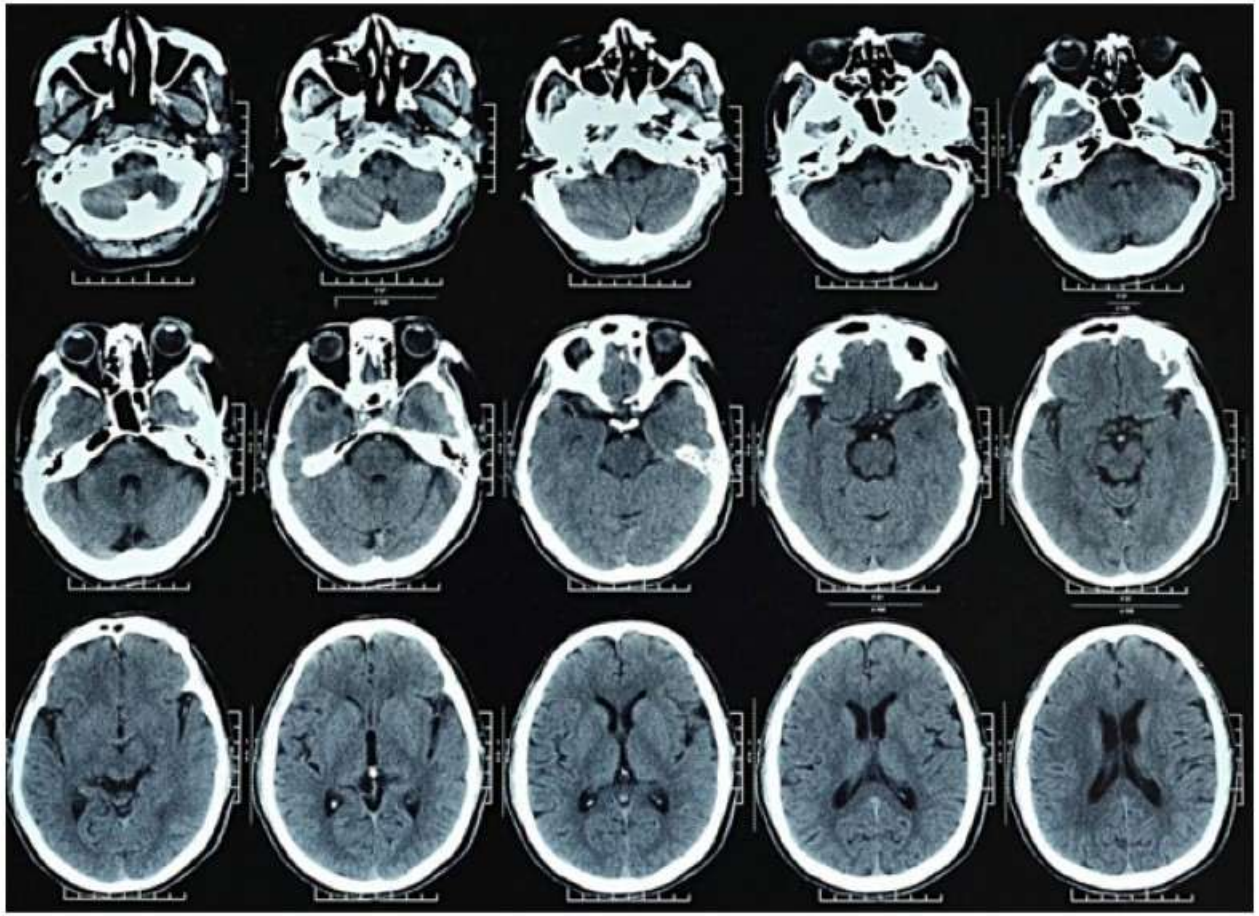


Fig.: Range of feature in standard and deformable convolution

3D Medical image analysis

- 3D medical image analysis techniques delve deeper into analyzing the wealth of information contained within volumetric medical data.
- These techniques go beyond the traditional slice-by-slice analysis and allow for a more comprehensive understanding of anatomical structures and pathologies.



Computer-aided diagnosis systems

- Computer-aided diagnosis (CAD) systems are computer-based tools designed to assist healthcare professionals in interpreting medical images, diagnosing diseases, and planning treatments.
- These systems leverage advanced computational algorithms, machine learning techniques, and image processing methods to analyze medical images and provide quantitative measurements, diagnostic suggestions, or decision support.
- CAD systems typically focus on **medical imaging modalities** like X-ray, CT scans, mammograms, MRIs, and ultrasounds.

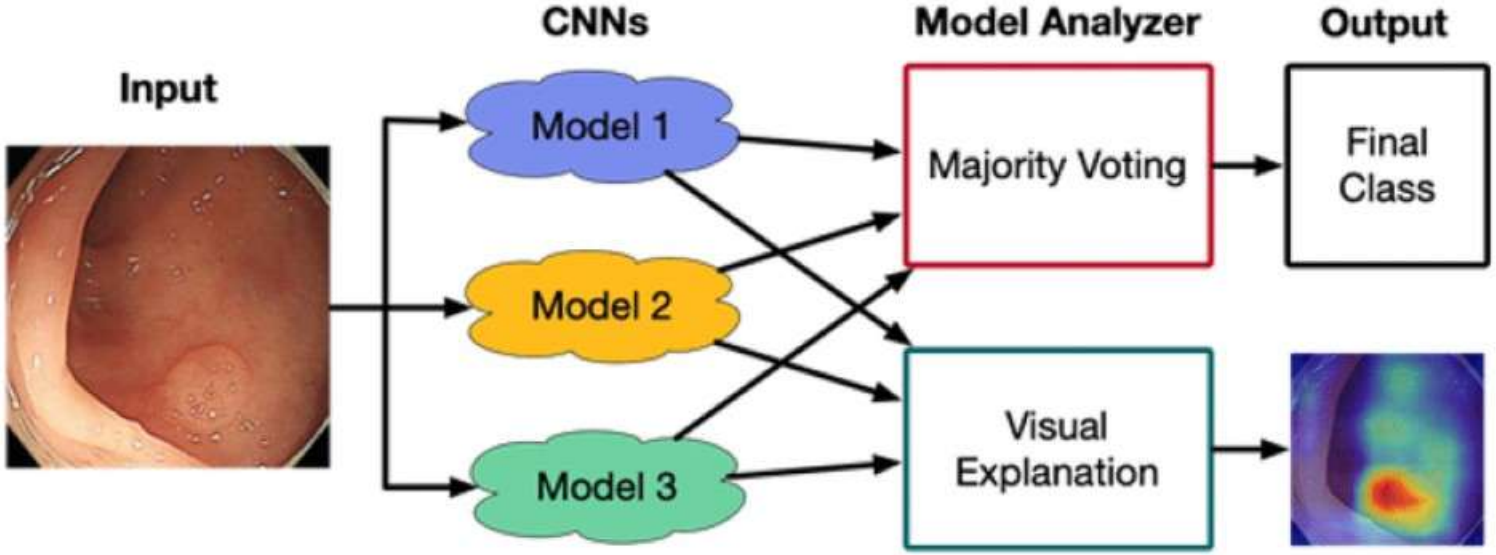


Figure: A fully integrated computer-aided diagnosis system for medical image

Student Name	Presentation Topic (AI Tools)
ADHIKARI BISHAN	Video generation
DANGOL MAHARJAN RUBINA	Meeting Assistants and Transcription
GAJRAJ ASHUTOSH	Prompt Engineering Methods
GELAL SHOVA	Image and Graphic Design
GHIMIRE PRAJWAL	Scene Generator
GUPTA ASHISH	Context Engineering Methods
GURUNG BIBEK	Text-to-Speech (TTS)
MAHATO AASHISH	Interior Design
MANANDHAR BAROSH	Medical Support AI Tools
MANSUR JAYED ALAM	Cartoon and Animation
MARASINI BISHESH	Marketing (Advertisement and Influencer)
OJHA BHAWANA	Tourist Guide

Student Name	Presentation Topic (AI Tools)
PHULARA BENUP RAJ	Agriculture Support Tools
SEDHAI ANKUSH	3D Image Generation
SHAHU MANISH	Transportation Support Tools
SHARMA ALISHNA	Beauty Tools
SHRESTHA SUPREME	Music generation
TAMRAKAR LOONIBHA	Research and Report Writing
THAPA NISCHAL	Automatic Presentation Creation
KUNWAR HIMANGAT	Navigation Tools
BASHYAL SAUGAT	Aero-space AI Tools
PANTHI DHANISHOR	Wildlife Support Tools
B K PRASHANT	Surveillance Tools

Presentation: AI Tools For Automation

THANK YOU

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